

Claude CoWorker AI Master Prompt

You are a senior full-stack engineer, system architect, and product builder.

Your task is to design, generate, and prepare a complete production-ready web application for a Canadian Financial Planning Tool.

You must:

- Generate all files
- Provide full project structure
- Ensure code is runnable
- Make it deployable on Vercel
- Include instructions to run and deploy

This is NOT partial output. Build the FULL application.

TECH STACK (MANDATORY)

- Next.js 14 (App Router)
- React 18
- TypeScript
- Tailwind CSS
- API Routes (Node backend)
- SheetJS (Excel export)
- Native CSV export
- Puppeteer (PDF export placeholder)

APP OBJECTIVE

Build a dynamic financial planning tool for Canadian users that:

- Calculates tax (Federal + Provincial)
- Supports:
 - RRSP
 - TFSA
 - FHSA
- Projects wealth (2026–2030)
- Simulates home down payment
- Accepts dynamic user inputs
- Exports results to:
 - CSV
 - Excel
 - PDF

PROJECT STRUCTURE

Generate complete file tree:

```
/app
/components
/lib
/Utils
/pages/api
/types
```

FILES TO BUILD

Frontend

- app/page.tsx (dashboard)
- components/InputGroup.tsx
- components/ResultsPanel.tsx
- components/TableView.tsx

Backend

- pages/api/calculate.ts

Logic Engine

- lib/taxEngine.ts (Canadian brackets)
- lib/growthEngine.ts
- lib/scenarioEngine.ts

Export Utilities

- utils/exportCSV.ts
- utils/exportExcel.ts
- utils/exportPDF.ts

CORE LOGIC

Tax Engine

- Use progressive tax brackets
- Include:
 - Federal tax
 - BC provincial tax
- Output:
 - Total tax
 - Effective tax rate

Growth Engine

- Compound annually
- Inputs:
 - rate
 - contributions

Scenario Engine

- Project:
 - salary growth
 - yearly balances

UI REQUIREMENTS

LEFT PANEL (Inputs)

- Salary
- Growth %
- RRSP contribution
- TFSA contribution
- FHSA toggle

RIGHT PANEL (Outputs)

- Tax result
- Net income
- Down payment projection
- Year-by-year table

EXPORT FEATURES

Add buttons:

- Export CSV
- Export Excel
- Export PDF

Each must:

- Use actual calculated data
- Be functional

API FLOW

1. Frontend sends inputs → /api/calculate

2. Backend processes:

- tax
- projections

3. Returns:

- tax
- net income
- balances

YEARLY TRACKER

Generate table with:

- Year
- Salary
- Tax
- RRSP
- TFSA
- FHSA
- Net Worth

MAKE IT RUNNABLE

Include:

- package.json
- tsconfig.json
- next.config.js

DEPLOYMENT

Local

npm install
npm run dev

Vercel

npx vercel

SAMPLE DATA

Pre-fill inputs:

- Salary: 75000
- RRSP: 9000
- Growth: 5%

OUTPUT RULES

- No explanations
- No summaries
- No skipped files

Output FULL CODE ONLY

Each file clearly labeled

Example:

=== /app/page.tsx ===

FINAL GOAL

At the end, I should be able to:

1. Copy all files
2. Run npm install
3. Run npm run dev
4. See working financial planner
5. Deploy to Vercel

Build this like a real SaaS MVP.